

## PROFILE

App and web developer building production Android and full-stack systems. Software developer at Kosal Tech Solutions, with a contract engagement at Manju Global — delivering Kotlin/Compose mobile apps, Next.js platforms, and the Go and Python services behind them. Computer Science Engineering student (2022–2026) focused on AI-powered applications and clean, maintainable architecture.

## EXPERIENCE

## Software Developer

Kosal Tech Solutions

CURRENT

- Build internal tooling and client-facing web systems across the company's delivery work.
- Developed Kosal-Link, a four-tool scraper suite — static downloader, browser-based scraper for JS-heavy pages, structure extractor, and a CLI that renders a live landing page, detects section boundaries, and exports selector-scoped components as a reusable blueprint.

## App &amp; Web Developer · Contract

Manju Global

CONTRACT

- Mconnect — Android client for the Manju Groups project-management system: OTP authentication, HR, attendance, leave and chat. Kotlin + MVVM + Retrofit, AES256-GCM encrypted session storage, and a semantic colour-token design system driving light and dark from one source.
- Manju PMS platform — Next.js + Convex operations platform covering land procurement, legal and feasibility gating, channel-partner management, Excel/PDF reporting and an AI sales assistant.
- Airix Geo Tracking — Go service for location ingestion, live tracking, trip sessions and tamper monitoring on PostgreSQL/PostGIS + TimescaleDB, with Redis for live state and idempotency.
- Travel Desk — Next.js trip allocation and fleet billing: token-scoped driver pages delivered over WhatsApp with no driver login, OCR on uploaded odometer evidence, and explicit extra-km, cancellation and offline-completion states.

## MERN Stack Developer Intern

VEI Technologies

JAN-FEB 2025

- Developed full-stack modules using MongoDB, Express.js, React and Node.js.
- Built and integrated REST APIs connecting frontend components with backend services across agile sprints.

## Web Development Intern

NoviTech

JAN 2025

- Designed responsive web interfaces in HTML, CSS and JavaScript, integrating backend services and optimising page performance across multiple modules.

## SELECTED PROJECTS

## Mentorix — AI Career Intelligence System

Python · FastAPI · RandomForest · SQLite · Vercel / Render

[github.com/D-A-R-X/mentorix-ai](https://github.com/D-A-R-X/mentorix-ai)

AI engine analysing student behaviour patterns to predict career instability and recommend personalised paths. Modular FastAPI REST services with a RandomForest model behind an explainable AI layer that returns reasoning, not just scores.

## EDUCATION

## B.E. Computer Science

Dhanalakshmi Srinivasan College of Engineering

Coimbatore, Tamil Nadu

2022 – 2026

CGPA 7.43 / 10

## TECHNICAL SKILLS

## LANGUAGES

Python, JavaScript, TypeScript, Kotlin, Dart, Go

## FRAMEWORKS

React, Next.js, FastAPI, Jetpack Compose, Flutter, Node.js

## MOBILE

Android (Kotlin / Compose), Material 3, Flutter

## BACKEND

Convex, Firebase, REST APIs

## DATA

PostgreSQL / PostGIS, TimescaleDB, SQLite, Firestore

## AI / ML

Machine Learning, Generative AI, Prompt Engineering

## TOOLING

Git, GitHub, Docker, VS Code, Cursor

## DEPLOYMENT

Vercel, Render

## CERTIFICATIONS

## ORACLE UNIVERSITY

OCI 2023 Certified Foundations Associate

Jun 2024 – Jun 2026

## NXTWAVE

AI for Students: Build Your Own Generative AI Model

Sep 2024

## GOOGLE DIGITAL UNLOCKED

Fundamentals of Digital Marketing

Jan 2021 · ID 6A5 5B9 RR5

## RECOGNITION

- Published 'AI in Retail Marketing' — PSG College Conference
- MSME Hackathon — Rathinam Institute of Technology
- 3+ National Hackathon presentations
- MERN Stack Workshop — CIT Coimbatore, Feb 2025

## INTERESTS

AI systems & applications, backend architecture, Android development, geospatial systems, AI product development, full-stack engineering.

**Darx Dialer** — Default Phone App with Cloud Recording

Kotlin · Jetpack Compose · InCallService · Convex

Full default-dialer replacement implementing InCallService and the DIALER role — lock-screen incoming UI, Bluetooth routing, DTMF and hold. Per-call recording is consent-first: on-screen indicator, persistent notification and a user toggle, never covert.

**Brownie Care** — AI Companion Health App

Kotlin · Compose Canvas · Groq LLM · Convex · WorkManager

Health companion built around a character drawn entirely in Compose Canvas with spring animations — no image assets. DataStore is the source of truth with Convex as a best-effort cloud mirror, so the app works fully offline.

**Megatron** — Modular Adaptive AI Control SystemPython · FastAPI · JavaScript [github.com/D-A-R-X/Megatron](https://github.com/D-A-R-X/Megatron)

Modular AI controller with a central reasoning layer that analyses intent and delegates to specialised modules — new capabilities added without rewriting core logic.

**Smart Campus** — Academic Resource Management App

Flutter · Firebase Auth · Firestore · Firebase Storage

[Play Store](#) — [com.schrodingerlab.curiosity](https://com.schrodingerlab.curiosity)

Cross-platform app delivering centralised academic resources with offline download support. Deployed to the Google Play Store.

**Drive Track Mate** — Intelligent Driving Monitoring

Android · Dash Camera Integration · Event-Based Recording

[github.com/D-A-R-X/drive-track-mate](https://github.com/D-A-R-X/drive-track-mate)

Event-aware recording that auto-tags sudden braking and abrupt manoeuvres with timestamps and metadata, replacing unstructured continuous footage with retrievable evidence.

**Fitcore** — AI-Based Fitness Tracking System — in progress

Flutter · Firebase Auth · Firestore

Adaptive fitness progression using verified activity data and experience-based levelling — progress calculated from validated signals, not user-controlled inputs.

## ALSO

**Cruza**

Co-founder · independent software studio  
building intelligent systems. Est. 2025.